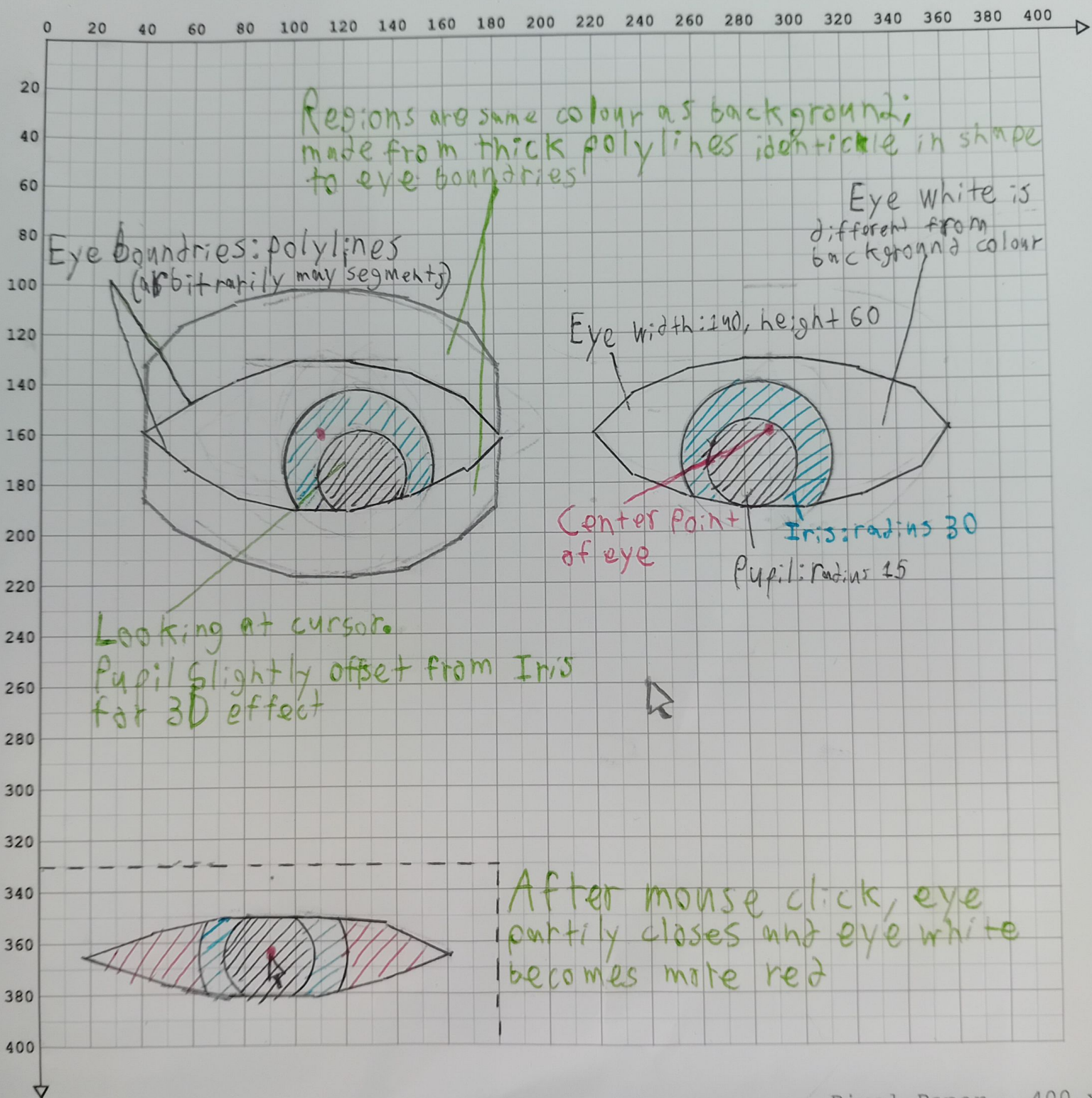




Assignment 2 Process Work

Pre-Planning:

The project is for there to be a pair of eyes that will look at the mouse and ~~move~~ become agitated when clicked.

I chose this idea for my project because this was my favorite out of the 5 and it felt the most creative and original. I initially thought it wouldn't be feasible until I had an idea for how it can be done. *

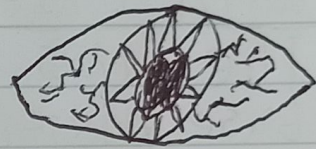
The irises and pupils will be circles. They will have base positions at the centres of the eyes. The final positions are ~~moved~~ shifted towards the mouse (I think I can use $\sin$ and $\cos$ for direction) an amount based on ~~the~~ a function of the distance to the mouse (function to implement). The pupils will be shifted slightly more to give the slight illusion of 3D; multiply by a small value, ex. 1.1.

* The boundaries of the eyes can be made using polylines. They can have as many segments as possible to increase smoothness. Above and below each eye, there will be more polylines that are the same colour as the background, the same shape as the boundaries but much thicker, and just barely touch the boundaries. If they are the top layer, they will cover the pupils and irises when they are past the edge of the boundaries, giving the illusion that they are going under the eyelids. It will also allow for the whites of the eyes to be simple shapes that can have their colour changed.

When the mouse clicks on an eye (hitbox may be crude), that eye will have its white turn slightly red, and the eye will partially close. I will store the agitation in a variable from 0 to 255, that way I can just directly plug it into the red value, and subtract ~~hit~~ ^{divided by} 8 from the y-values of the points in the polylines. I can decrease the agitation value every frame to slowly fade the effects.

Optional future features:

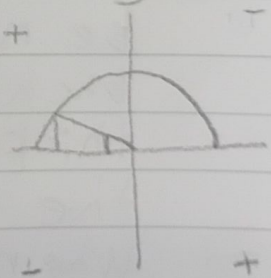
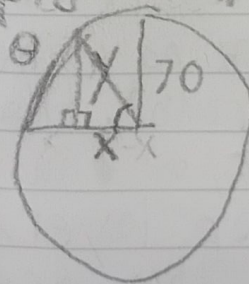
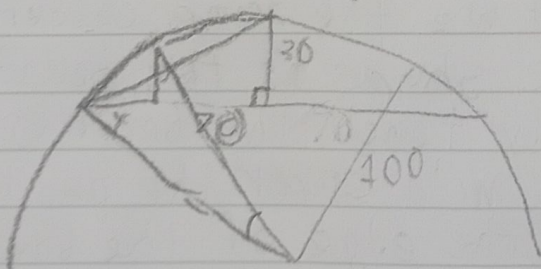
- The eyes blink
- Add detail to the irises
- The eyes make tears
- Add veins to the eye whites



Log ~~1~~ 1:

I added a circle of radius 100 to help align the polylines later. I had to try different positions to get it right.

my bad attempts at working out



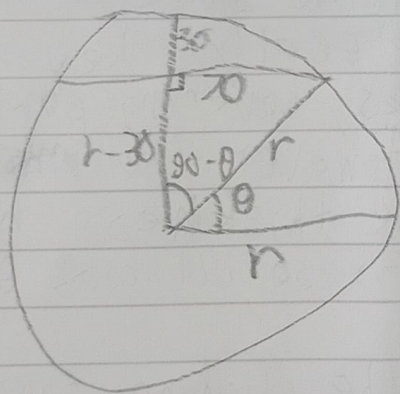
Log 2:

I'm not allowed to use polylines, so I have to draw individual lines instead.

To get the points of the lines to ~~map~~^p onto a circle, I can subtract from the center points by 70, scaled using sin and cos. I can then change the vertical number to 30 to get an eye shape. Initially, I tried to use tan for this for some reason. Later, I can multiply these terms by an agitation scale value (0 to 1).

However, I decided this would look too much like an ellipse; the way to get the sharp angles that eyes have is to use 2 intersecting circles. If I use different center points, larger values, and start at a higher angle, I can get the look I want.

working out the size of the circle and starting angle



$$r^2 = (r-70)^2 + 30^2$$

$$r^2 - 60r + 900 + 4900 - r^2 = 0$$

$$r = 96 \frac{2}{3}$$

$$\sin(\theta) = \frac{70}{r} \quad 90 - \theta = 46.397^\circ$$

$$\theta = 43.6^\circ$$

Log 3:

Got that working; the bottom boundary is identical but with addition and a different center point. It took a minute to adjust. I went with 20 line segments.

Log 4:

I added the pupil, iris, and eye white, along with the movement. I first tried to shift the x and y values of the circles by adding the square root of the distances to the mouse, respectively.

* also, the pupil's displacement is multiplied by 1.3

this resulted in the eye moving faster when the mouse was closer (good), but separately ~~based~~ ~~on~~ in x and y based on which distance was closer; the eye was not aligned to the mouse. Now, it uses the square root of the direct distance for the total displacement, and scales it in x and y with $\sin$ and $\cos$. I had some difficulty with the y direction because $\tan$ (which I had to use to get the angle) only goes from -90 to 90° , but I eventually got it working with some trial and error (playing with abs and sign).*

Log 5:

I added the 'eyelids', which are identical to the boundaries, except for that they are thicker, ~~are~~ shifted above and below their boundary counterparts, and they are the same colour as the background. If aligned perfectly, they should wrap around the boundaries, covering parts of the iris that go outside. However, the ~~inner~~ curve changes shape as the line thickness increases, ruining it. I can't find a fix so I'll come back to this problem later.

Log 6: I ~~have~~ added the clicking functionality. The agitation variable (will be separate for each eye), starts at 0, and increases to 0.2 when the eye is clicked, up to a maximum of 1. It decreases by 0.0025 every frame, taking 80 frames to remove the effects of one click.

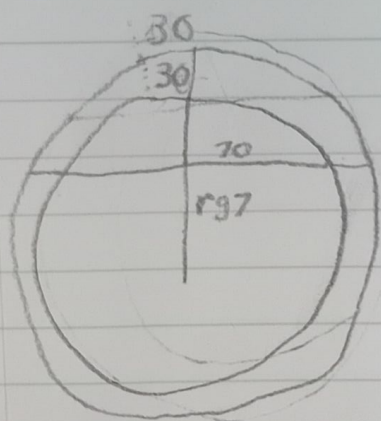
I added a term to the y positions of the points of the lines: it is the difference between the base position and 160 (middle y of eyes), times 0.75 (so they only mostly close), scaled by the agitation value.

I had tried a different method before this coming up with this one, involving the ratio between the angle use for the first point and the one used at each point, but this caused some points to move too much and others not enough.

I should be able to just multiply the blue and green values of the 'eye white' colour by the agitation to get the redening effect, but SetFillColor does not accept float values.

Log 7:

The multiplied value is type cast into an int. I realized that multiplying by agitation directly would make it start red and turn white, so I need to use 1-agitation.



$$\sin(90 - \theta) = \frac{100}{227} \quad \theta = 31.27$$
$$\Delta \theta = 11.746$$

Log 8:

I tweaked the parameters of the 'eyelids'. They are not perfect, but good enough.

The solution to the earlier problem was to use a bigger radius instead of shifting them.

Log 9:

I added the right eye. After changing the x coordinates, it works and is in the right place. I also had to remove a couple of lines that overlapped the other eye.

Project complete.